

Program : Diploma in Polymer Technology	
Course Code : 4091	Course Title: Polymer Processing & Machinery
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To understand the different mixing equipments and mixing processes used in rubber industry.
- To understand machinery parts of different mixers, moulding equipments, extruder, calendar and thermoforming machine.
- To acquire the knowledge of different processing methods for the polymer materials.

Course Prerequisites:

Topic	Course code	Course name	Semester
Classification of polymers		Fundamentals of polymer science	2
Basic electrical engineering		Fundamentals of Electrical and Electronics Engineering	1
Basic mechanical engineering		Basic mechanical engineering	3
Rubber compounding		Technology of Elastomers	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Describe the mixing machinery and process.	12	Understanding

CO2	Explain the moulding process and machinery.	17	Applying
CO3	Explain the working principles, parts and accessories of extruders and blow moulding machinery	17	Applying
CO4	Paraphrase rotational moulding, calendars, thermoforming and auxiliary equipments	12	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	3					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Describe the mixing machinery and process.		
M1.01	Classify different mixing equipments	1	Remembering
M1.02	Explain the working principle, parts and mixing sequences of two roll.	3	Understanding
M1.03	Explain the working principle, parts and mixing sequences of internal mixers.	4	Understanding
M1.04	Illustrate Continuous mixing equipments and plastic compounding equipments.	4	Understanding
Contents: Mixing machinery Classification- batch and continuous. Working principle- parts-accessories, safety devices-uses of two roll mill and internal mixers. Differentiate Banbury and Intermix. Fill factor. Mixing sequences-distributive mixing, dispersive mixing- upside down mixing. Continuous mixing - Farrel continuous mixer. Auxiliary mixing equipments used for plastic compounding.			

CO2	Explain the moulding process and machinery.		
M2.01	Discuss the compression and transfer moulding machine and process.	6	Understanding
M2.02	Explain the injection moulding machine and process	7	Understanding
M2.03	Identify different defects and Troubleshooting of I/M, C/M and T/M process	2	Applying
M2.04	Describe RIM and RRIM	2	Understanding
	Series Test – I	1	
Contents: Compression moulding - process - moulds - flash, positive - semi positive. Hydraulic press - working principle-parts-accessories. Transfer moulding - process, machinery. Injection moulding- classifications –for rubbers and plastics, ram & reciprocating screw type. Reciprocating screw type - parts - accessories - moulds - clamping system-ejection mechanism. Differentiate rubber and plastic injection moulding.			
CO3	Generalize the working principles, parts and accessories of extruders and blow moulding machinery		
M3.01	Illustrate the parts and accessories of extruders.	5	Understanding
M3.02	Illustrate working process of extruders.	5	Understanding
M3.03	Categorize different types of extruders.	3	Understanding
M3.04	Explain blow moulding process	4	Understanding
Contents: Extrusion - Definition - process. Extruder – classifications - extruder size - cold feed - hot feed - T - head - dual - twin screw - vented. Extruder –screw type - parts, accessories – blown film extrusion. Blow moulding - extrusion blow moulding- parison programming - injection blow moulding.			
CO4	Paraphrase rotational moulding, calendars, thermoforming and auxiliary equipments		
M4.01	Illustrate calendaring process and machinery	4	Understanding
M4.02	Describe various thermoforming process	4	Understanding
M4.03	Explain rotational moulding process	4	Understanding
	Series Test – II	1	

Contents:

Calendar - configuration, parts, accessories. Roll deflection and compensation technique like roll cambering (crowing) roll crossing and roll bending. Temperature control system of calendars. Thermoforming - vacuum forming, pressure forming, and plug assisted forming - products - examples. Rotational moulding - process.

Text / Reference:

T/R	Book Title/Author
T1	Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth Hainemann Ltd, 1982)
R1	Rubber processing and production organization -Philip K Freakley (Publisher: Springer 2011).
R2	Blow moulding -David A Jenners
R3	Injection moulding -Irvin Ruhin
R4	Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing Co. 1986)

Online Resources:

Sl.No	Website Link
1	www.researchgate.net
2	www.scribd.com
3	www.iri.net.in
4	www.sciencedirect.com
5	www.timcorubber.com